

Bugs found: 4

1. Accesses `arr[3]` in a 3-element array, causing `ArrayIndexOutOfBoundsException`.
2. `1 << 31` sets the sign bit and produces a negative `int` value due to integer overflow/sign overflow.
3. `Bad.equals` always returns `false`, violating the `equals` contract, including reflexivity.
4. `Bad` overrides `equals` but does not override `hashCode`, violating the `equals/hashCode` contract for hash-based collections.